TOPIC 03 · FINANCIAL MANAGEMENT (I)

Financial Management (I): Financial Statements and Financial Structure

Economic and financial structure, the income statement, the balance sheet, the sources of financing, the debt-to-equity ratio and the weighted average cost of capital.

Source: Universidad Carlos III de Madrid · Topic 3_Financial Management (I).pdf (Prof. Carmen Paz-Aparicio), Slides 1-37

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Central idea: a firm has two sides. Its assets (the economic structure, what it owns) generate profitability; its equity and liabilities (the financial structure, what it owes) carry a cost. Value is created when profitability exceeds cost. Two statements describe the firm: the income statement, which measures performance over a period (revenues − expenses = net income), and the balance sheet, which shows the position at one moment (assets = liabilities + equity). The financial structure combines self-financing, external equity and long- and short-term debt; it is measured with the debt-to-equity ratio and priced with the weighted average cost of capital.

The topic has three parts [Slide 2]: an introduction to the economic and the financial structure; the financial statements, namely the income statement and the balance sheet; and the financial structure of the company.

Part I — Economic and financial structure

§ 1 The finance area and its two decisions [Slide 3]

The **finance area** of a company is in charge of the **collection and management of the funds (capital)** the company uses. The finance manager faces two types of decision: what to invest in and when (**investing decisions**), and how to obtain the money to invest (**financing decisions**).

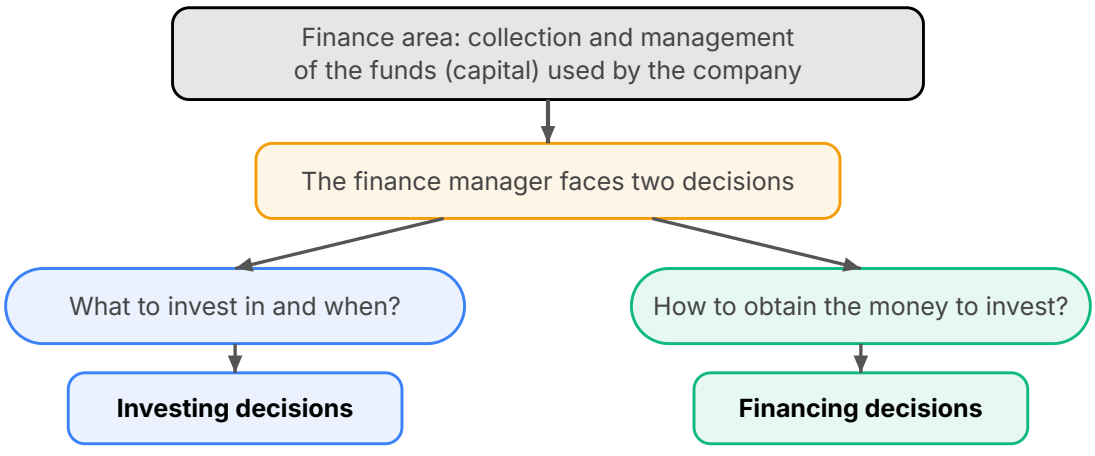


Figure 1. The finance manager's two questions and the decisions they correspond to.

§ 2 Economic structure and financial structure [Slides 4-5]

The **economic structure** is defined by the **investments** the company carries out. It represents the use of the financial resources in the form of goods (assets) and rights: the **assets**, or what is owned.

The **financial structure** is defined by the **sources of financing**. It includes all the financial resources used to carry out the investments (debt and liabilities): the **liabilities**, or what is owed [Slide 4].

The assets produce **profitability**, while equity and liabilities have a **cost**. The objective of the company is to **maximize value**, which requires **profitability > cost** [Slide 5].

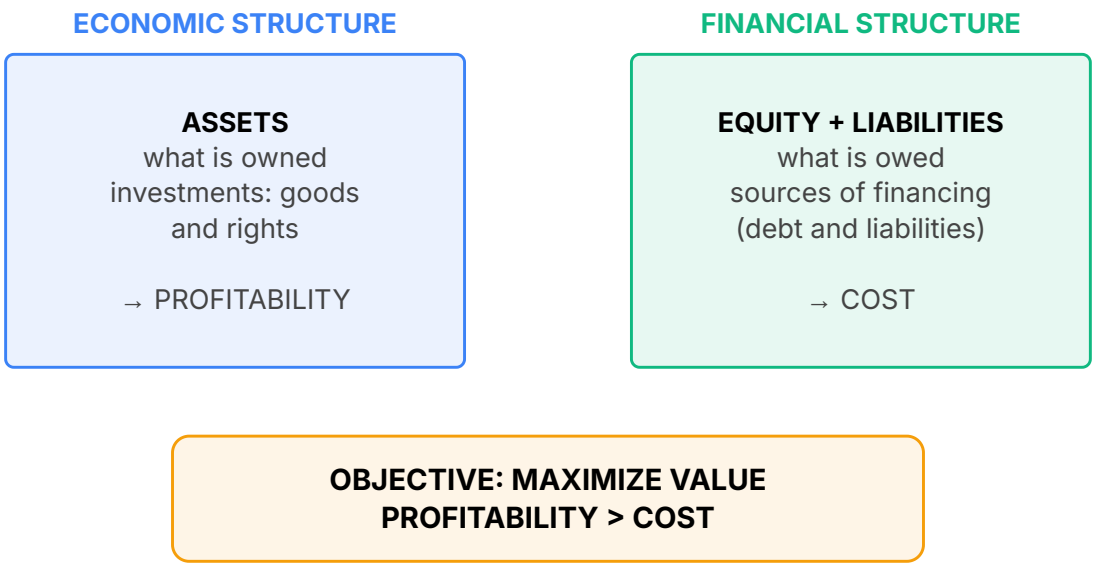


Figure 2. The two sides of the firm: the assets (economic structure) generate profitability, equity and liabilities (financial structure) carry a cost; value is created when profitability exceeds cost.

§ 3 What the company's decisions focus on [Slide 6]

Financing decisions concern the choice between **equity and debt**. The financial resources available can be **internal equity** (self-financing, i.e. retained earnings), **external equity** or **debt**. Financing decisions involve decisions on the **capital structure** and on **dividends**.

Investing decisions consist in allocating capital to the investment projects from which future profits are expected. Companies invest in the projects that **create value** for them, those in which **profits > costs**.

Part II — Financial statements

§ 4 The income statement [Slides 7-8]

The **income statement** is the first statement prepared. It reports the **performance** of the business **over a specific period**, and in its simplest form [Slide 7]:

$$\text{Revenues} - \text{Expenses} = \text{Net income}$$

Revenues are earned from the sale of goods or services and are recorded **when the sale is made**, whether or not the payment has been received; examples are sales, service revenue and interest revenue. **Expenses** are incurred **when the business receives goods and services**, whether or not it has paid for them; examples are salaries, utility and interest expense [Slide 8].

§ 5 The income statement: extended format [Slides 9-11]

Line	
Sales revenue	
− Cost of goods sold	
= Gross profit	
− Operating expenses	
= Income from operations	
± Non-operating items	
= Income before taxes	
− Income taxes	
= Net income	

The lines are defined as follows:

- The **cost of goods sold** is the expense the business incurred to buy or make a product for resale. If a bookshop buys a book for 25 USD and sells it for 32 USD, the cost of goods sold is 25 USD [Slide 10].
- **Operating expenses** are the usual expenses of running the business, such as salaries, utilities and depreciation [Slide 10].
- **Non-operating items** are the revenues, expenses, gains and losses that do not relate to the company's primary operations, such as interest expense and gains and losses on the sale of equipment and investments [Slide 11].
- **Income taxes** are the income before taxes multiplied by the tax rate [Slide 11]. With an income before taxes of 50,000 USD and a rate of 30%:

$$\text{Income taxes} = 50{,}000 \times 0.30 = 15{,}000 \text{ USD}$$

§ 6 The balance sheet [Slides 12-13]

The **balance sheet** reports the **financial position** of an accounting entity **at a particular point in time** [Slide 12]:

$$\text{Assets} = \text{Liabilities} + \text{Equity}$$

- **Assets** are the economic resources owned by the company: cash, accounts receivable, supplies, buildings, equipment.
- **Liabilities** are the company's debts or obligations: accounts payable, unearned revenues, bonds payable.
- **Equity** is the residual balance, $\text{Assets} - \text{Liabilities} = \text{Equity}$. In a corporation it is called **stockholders' equity**, because it represents the financing provided by the stockholders together with the earnings not paid out as dividends (**retained earnings**) [Slide 13].

§ 7 Classes of assets, liabilities and equity [Slides 14-17]

Assets are split into **current** and **non-current** [Slides 14-15]:

$$\text{Current assets} + \text{Non-current assets} = \text{Total assets}$$

- **Current assets** will be used or turned into cash **within one year**: cash, accounts receivable, inventory, short-term investments, supplies and prepaids.
- **Non-current assets** are all the others: long-term investments, land, buildings, equipment and patents.

Liabilities are split into **current** and **long-term** [Slide 16]:

$$\text{Current liabilities} + \text{Long-term liabilities} = \text{Total liabilities}$$

- **Current liabilities** will be paid in cash (or other services) or satisfied **within the coming year**.
- **Long-term liabilities** will not be paid or satisfied within the year.

Stockholders' equity is split into **contributed capital** and **retained earnings** [Slide 17]:

$$\text{Contributed capital} + \text{Retained earnings} = \text{Total stockholders' equity}$$

§ 8 The complete balance sheet [Slide 18]

$$\text{Current assets} + \text{Non-current assets} = \text{Current liabilities} + \text{Long-term liabilities} + \text{Stockholders' equity}$$

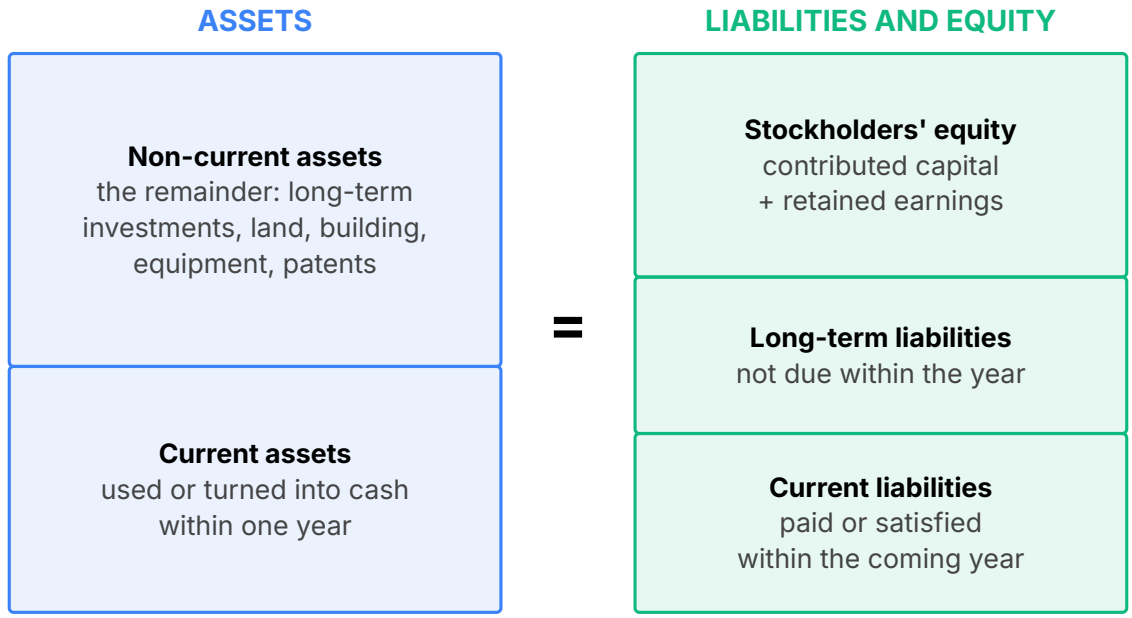


Figure 3. Layout of the balance sheet: total assets (non-current + current) equal total liabilities and stockholders' equity. Block sizes are only illustrative.

§ 9 Worked example: from a list of accounts to the two statements [Slide 19]

A company has the following accounts (tax rate 30%).

Step 1. Classify each account.

Account	Amount	Class	Position
Cash	5,000	Asset	Current asset
Accounts receivable	10,000	Asset	Current asset
Inventories	45,000	Asset	Current asset
Supplies	4,000	Asset	Current asset
Buildings	65,000	Asset	Non-current asset
Accounts payable	12,000	Liability	Current liability
Bonds payable	40,000	Liability	Long-term liability [Slide 29]
Common stock	45,000	Equity	Contributed capital
Additional paid-in capital	20,000	Equity	Contributed capital
Retained earnings (beginning balance)	5,000	Equity	Retained earnings
Sales	100,000	Revenue	Sales revenue
Cost of goods sold	58,000	Expense	Cost of goods sold
Utility expense	8,000	Expense	Operating expense
Supplies expense	3,000	Expense	Operating expense
Salaries expense	16,000	Expense	Operating expense
Interest expense	5,000	Expense	Non-operating item

Supplies (an asset still held) and **supplies expense** (the part consumed in the period) are different accounts.

Step 2. Income statement.

Line	Computation	Amount
Sales revenue		100,000
− Cost of goods sold		58,000
= Gross profit	$100{,}000 - 58{,}000$	**42,000**
− Operating expenses	$8{,}000 + 3{,}000 + 16{,}000$	27,000
= Income from operations	$42{,}000 - 27{,}000$	**15,000**
− Interest expense		5,000
= Income before taxes	$15{,}000 - 5{,}000$	**10,000**
− Income taxes (30%)	$0.30 \times 10{,}000$	3,000
= Net income	$10{,}000 - 3{,}000$	**7,000**

Step 3. Retained earnings at the end. With no dividends, $5{,}000 + 7{,}000 = 12{,}000$.

Step 4. Balance sheet.

Assets	Amount	Liabilities and stockholders' equity	Amount
Cash	5,000	Accounts payable	12,000
Accounts receivable	10,000	**Current liabilities**	**12,000**
Inventories	45,000	Bonds payable	40,000
Supplies	4,000	**Long-term liabilities**	**40,000**
Current assets	**64,000**	Common stock	45,000
Buildings	65,000	Additional paid-in capital	20,000
Non-current assets	**65,000**	Retained earnings	12,000
		Stockholders' equity	**77,000**
Total assets	**129,000**	**Total liabilities and equity**	**129,000**

Check: $129{,}000 = 12{,}000 + 40{,}000 + 77{,}000$. The balance closes with no tax liability, so the cash of 5,000 is the cash after the 3,000 of taxes have been paid.

Part III — The financial structure of the company

§ 10 Financing decisions and the capital structure [Slide 20]

Financing decisions analyse the different financial alternatives. **Business financing** is the collection of the financial resources needed to carry out the company's activity, and the combination of all the possible financial sources is the **capital structure** or **financial structure**.

§ 11 The map of the financial structure [Slides 21-22]

The sources of financing can be classified by their **nature**, equity or liabilities [Slide 21], or by their **origin**, the company's own resources or external resources [Slide 22]. Both classifications draw the line in the same place:

Nature	Origin	Instruments
Equity	**Company resources**	Internal equity: self-financing
		External equity: equity capital and shares
Liabilities	**External resources**	Long term: bonds, long-term loans and bank credits, leasing
		Short term: accounts payable, trade discount, short-term loans and bank credits, factoring

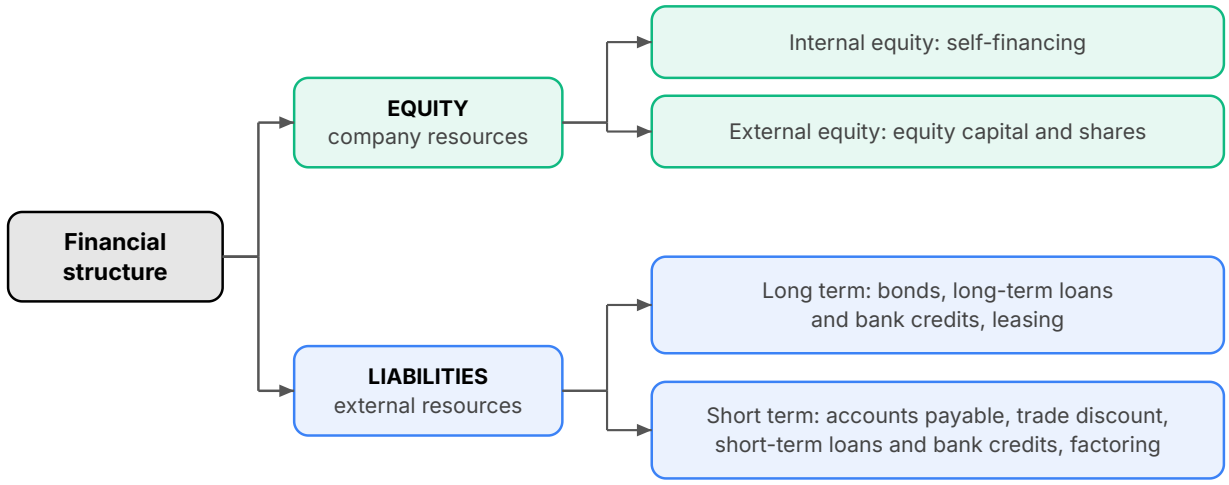


Figure 4. The financial structure: equity (the company's own resources, internal or external) and liabilities (external resources, long term or short term), with the instruments of each.

§ 12 Self-financing [Slides 23-24]

The **retained earnings**, the resources the company generates in its normal activity, constitute its **self-financing**. Profits can either be **retained** in the company (self-financing) or **distributed** (dividends). Two ratios describe the split:

$$\text{Pay-out} = \frac{\text{Dividends}}{\text{Profits}}, \qquad \text{Self-financing } (\%) = \frac{\text{Retained earnings}}{\text{Profits}}$$

With profits of 100,000 €, dividends of 75,000 € and retained earnings of 25,000 €, the pay-out is $75{,}000/100{,}000 = 75\%$ and the self-financing ratio is $25{,}000/100{,}000 = 25\%$ [Slide 23].

§ 13 External equity [Slides 25-26]

When a business is created, its owners put in funds to finance the assets. **Equity capital** is the money raised from the owners at the start of the business (**initial capital**) and through later **rights issues** (issues of new shares), used to finance the company's investments. It is **not repaid** to the investor at any maturity date, and a share can be turned into cash **only by selling or transferring it** [Slide 25].

A **share** is one of a finite number of equal portions of the capital of a company. It entitles its owner to a proportion of the distributed, non-reinvested profits (**dividends**) and to a portion of the value of the company in case of liquidation. Shares are equity securities and have several values: **nominal, accounting (book), intrinsic and market** value. New shares are issued through a **rights issue** (capital increase), and paying dividends is **not compulsory** [Slide 26].

§ 14 Bonds [Slide 29]

A **bond** is a **debt security**: the issuer owes the holders a debt and must repay the **principal** and pay **interest (the coupon)** at a later date, the **maturity**. It is a loan in the form of a security, in which the issuer is the borrower, the bondholder is the lender and the coupon is the interest. Bonds allow the issuer to finance **long-term investments**

with external funds.

§ 15 Long-term loans and bank credits [Slide 31]

With a **long-term loan**, the company receives an amount of money that must be paid back at a certain time, and pays for this service through the **debt interest**. With a **bank credit**, the financial institution makes a **maximum amount** available to the customer, who can borrow any amount up to that limit and pays interest **only on the amount borrowed**; interest rates are higher.

§ 16 Leasing [Slide 33]

Leasing lets a firm obtain the **use of a fixed asset** in exchange for a series of contractual, periodic payments. The **lessee**, who receives the asset or the services under the lease contract, has the **right to buy the asset** at the end of the agreed period. There are two types: **finance leasing** and **operating leasing**.

§ 17 Short-term external resources [Slide 35]

- **Accounts payable (trade credit):** financing from suppliers that arises when goods are paid for some time after they are received; suppliers allow the company to defer the payment.
- **Bank financing:** short-term **loans and credits**, with the same features as the long-term ones, and **trade discounts**, in which the financial institution advances to the customer the amount of a note (bill of exchange, promissory note, etc.) before it falls due, after deducting some expenses (interest, commission, etc.).
- **Factoring:** the company sells a credit it holds against a customer to a factoring company before it expires; the factoring company collects it and assumes the risk of insolvency.

§ 18 Measuring the financial structure: leverage [Slide 36]

The financial structure is the combination of the different financial resources. With E for equity and L for liabilities:

$$\%\,\text{Equity} = \frac{E}{E+L}, \qquad \%\,\text{Liabilities} = \frac{L}{E+L}, \qquad \text{Debt-to-equity ratio} = \frac{L}{E}$$

The **debt-to-equity ratio** evaluates the company's **financial leverage**: the degree to which it finances its operations through debt rather than its own funds. It reflects the ability of shareholders' equity to cover all outstanding debts in a business downturn, and **higher leverage ratios tend to indicate higher risk for shareholders**.

With total liabilities and equity of 1,000 M€, of which 700 M€ are equity and 300 M€ liabilities:

$$\%\,\text{Equity} = \frac{700}{1{,}000} = 70\%, \qquad \%\,\text{Liabilities} = \frac{300}{1{,}000} = 30\%, \qquad \frac{L}{E} = \frac{300}{700} = 0.43$$

§ 19 The cost of capital [Slide 37]

Every financial resource has a cost: **equity** has an **implicit (opportunity) cost**, and **liabilities** an **explicit cost**. The **capital cost** K is the weighted average of the costs of all the financial resources the company obtains:

$$K_{WACC} = K_{EQ} \cdot \frac{E}{E+L} + K_L \cdot \frac{L}{E+L}$$

where K_{EQ} is the cost of equity and K_L the cost of liabilities, weighted by the shares of equity and liabilities in the financial structure. With the structure of the leverage example (70% equity, 30% liabilities), $K_{WACC} = 0.70\,K_{EQ} + 0.30\,K_L$.

> Solved exercises for this topic (sheet exercises 1-6 and the four additional exam-type problems) →

Study notes written from the official course material (Topic 3 Financial Management (I).pdf; T3_Financial Statements_EXERCISES.pdf). Online version: subjects/business-management/teoria/topic-3.html